KHOA NGUYEN

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EDUCATION

New Jersey Institute of Technology September 2023 - Present

Ph.D. in Data Science

Focus: LLMs, VLMs, Post-Training (SFT & RL), Tool-Augmented Agentic Systems, AI for Software Engineer, Security, Privacy.

TECHNICAL SKILLS

- **Machine Learning:** LLMs, VLMs, Diffusion Models, NLP, RL, Distributed Training, GNN, Federated Learning, Distillation.
- **Frameworks:** Pytorch, Transformers, Llama-Factory, verl, vllm, Ray, Triton, DeepSpeed, Accelerate.
- **Infrastructures & Tools:** CUDA, Linux, Slurm, Git, Docker, SQL.
- **Programming Languages:** Python, Java, JavaScript, C/C++.

EXPERIENCE

AT&T | *Research Collaborator* November 2025 – Present

- **Cellular Network Maintenance Scheduling**: Designed a graph foundation model for LTE/5G site-maintenance prediction using large-scale historical network data, optimizing scheduling efficiency and user experience.

OppyAI Inc. | *Research Scientist* November 2024 – Present

- **Multi-Modal & Agentic Systems**: Extended a privacy-preserving split-LLM framework into a production enterprise solution, scaling capabilities from code generation to general chat, multilingual natural language understanding, VLMs, Tool-Augmented Agentic Systems, and MoE architectures.
- **RL for Reasoning Performance Optimization**: Integrated a distributed GRPO training loop into the post-training pipeline, significantly boosting the mathematical, logical, and structural reasoning capacities.
- **Adversarial Security & Robustness**: Conducted vulnerability assessments to design advanced attacks and built an adversarial training pipeline to defend models against privacy exploits.

New Jersey Institute of Technology | *Research Assistant* September 2023 – Present

- **Privacy-Preserving LLM System**: Developed `NOIR`, a distributed LLM system using differential privacy to prevent **100%** prompt and response reconstruction attacks. Built an end-to-end data pipeline to clean, refine, and transform raw pre-train code data into high-quality data optimized for large-scale SFT. Engineered a multi-GPU and multi-node distributed training pipeline using LLama-Factory, DeepSpeed, Accelerate, on an HPC Slurm cluster, achieving **4×** speedup.
- **Structure-Aware Code Intelligence**: Developed `GLMTest`, a graph-augmented LLM framework that leverages Code Property Graphs and semantic program analysis to improve LLM capabilities on generating test-cases toward targeted execution branches, achieving **71.9%** higher targeted-branch accuracy, **29%** higher branch overlap, and **2×** higher executable-test generation performance than frontier LLM baselines. Curated **3×** more branch-annotated dataset from real-world repositories by executing developer-written tests and aligning branch traces with program structure.
- **Federated LLMs for Chip Design**: Developed `FedChip`, a federated PEFT framework leveraging LoRA to collaboratively adapt language models for Verilog-based AI accelerator design under PPA constraints. Curated `APTPU-Gen` (30k design configurations) and introduced *Chip@k*, a statistical evaluation metric for multi-objective code quality, boosting optimal hardware generation by **77%** over frontier LLM baselines without leaking proprietary intellectual property.
- **Geographic FL**: Developed `SGFusion`, a stochastic FL algorithm that models correlations among geographic zones using hierarchical random graphs and MCMC sampling, enabling similar zones to share gradients through attention-weighted fusion; boosted mobile-sensing utility by **3.23%** across **2×** more zones without slowing convergence.

PUBLICATIONS

Papers

[1] K. Tran, **Nguyen, Khoa**, C. Borcea, et al., "Program Structure-aware Language Models: Targeted Software Testing beyond Textual Semantics," in *Findings of the Association for Computational Linguistics:* ***ACL'26***, 2026.

[2] **Nguyen, Khoa**, K. Ton, N. Phan, et al., "NOIR: Privacy-Preserving Generation of Code with Open-Source LLMs," in *Proceedings of the 35th USENIX Security Symposium*, ser. **USENIX Security '26**, 2026.

[3] M. Nazzal, **Nguyen, Khoa**, D. Vungarala, et al., "FedChip: Federated LLM for Artificial Intelligence Accelerator Chip Design," in *2025 IEEE International Conference on LLM-Aided Design (**ICLAD'25**)*, IEEE, 2025.

[4] **Nguyen, Khoa**, K. Tran, N. Phan, et al., "SGFusion: Stochastic Geographic Gradient Fusion in Federated Learning," in *2025 IEEE International Conference on Big Data (**BigData'25**)*, 2025.

Patents

[1] N. Phan, **Nguyen, Khoa**, K. Ton, et al., *System and Method for Private Generation of Code*, International Patent Application, No. PCT/US25/042391, 2025.

TEACHING EXPERIENCE

- Teaching Assistant: Basic Foundations of AI (FA25), Deep Learning (FA24, SP25), Programming Language Concepts (SP24)

ACHIEVEMENTS

- **Winner** – GfK's NextGen Data Science Hackathon (2022)
- **ICPC** – Qualified for Greater NY Regional Contest (2022)